Python Activity-Based Learning
Program 7: Retail Billing with Multiple Discount Rules

Problem Statement

Prepare a retail invoice for multiple products by calculating quantity-based prices, category-specific taxes, threshold discounts, and a maximum concession limit.

Objectives

• Calculate item-wise and subtotal amounts.

• Apply a discount based on the purchase total.

• Calculate category-specific tax and final invoice amount.

Assumptions

• Electronics tax is 18%; clothing tax is 5%; all other categories are taxed at 12%.

• A 10% discount is given when the pre-tax subtotal is at least ■5,000.

• Maximum discount is ■1,000. Tax is calculated after discount.

Algorithm

1. Read the number of products.

2. For each product, read name, category, unit price, and quantity.

3. Calculate each item amount and subtotal.

4. Calculate discount if subtotal reaches ■5,000, limited to ■1,000.

5. Calculate tax based on each product category after applying a proportional discount.

6. Display item details, discount, tax, and final total.

Flowchart

```
START → Read product count → Read each product → Calculate item amounts and subtotal → Check discount threshol
```

Pseudocode

```
BEGIN
  READ n
  subtotal = 0
  STORE each product amount and category
  FOR each product
    amount = price * quantity
    subtotal = subtotal + amount
  END FOR
  IF subtotal >= 5000 THEN discount = minimum of 10% subtotal and 1000
  ELSE discount = 0
  discount_rate = discount / subtotal
  FOR each product
    taxable_amount = amount * (1 - discount_rate)
    SELECT tax rate by category
    tax = taxable_amount * tax_rate
  END FOR
  DISPLAY subtotal, discount, tax, final total
END
```

Python Source Code

```
products = []
n = int(input("Enter number of products: "))
subtotal = 0

for i in range(n):
    print("\nProduct", i + 1)
    name = input("Product name: ")
```

```
    category = input("Category (electronics/clothing/other): ").lower()
    price = float(input("Unit price: "))
    quantity = int(input("Quantity: "))

    if price < 0 or quantity < 0:
        print("Invalid price or quantity.")
        continue

    amount = price * quantity
    subtotal = subtotal + amount
    products.append((name, category, amount))

if subtotal >= 5000:
    discount = min(subtotal * 0.10, 1000)
else:
    discount = 0

discount_rate = discount / subtotal if subtotal > 0 else 0
total_tax = 0

print("\nRETAIL INVOICE")
for product in products:
    name, category, amount = product
    taxable_amount = amount * (1 - discount_rate)

    if category == "electronics":
        tax_rate = 0.18
    elif category == "clothing":
        tax_rate = 0.05
    else:
        tax_rate = 0.12

    tax = taxable_amount * tax_rate
    total_tax = total_tax + tax
    print(name, "Amount:", round(amount, 2),
          "Tax:", round(tax, 2))

final_total = subtotal - discount + total_tax
print("Subtotal: Rs.", round(subtotal, 2))
print("Discount: Rs.", round(discount, 2))
print("Total tax: Rs.", round(total_tax, 2))
print("Final amount: Rs.", round(final_total, 2))
```

Test Case

Product 1: Phone, electronics, ■4,000, quantity 1. Product 2: Shirt, clothing, ■2,000, quantity 1.
Expected: subtotal ■6,000; discount ■600; tax ■738; final amount ■6,138.

Sample Output

```
RETAIL INVOICE
Phone Amount: 4000.0 Tax: 648.0
Shirt Amount: 2000.0 Tax: 90.0
Subtotal: Rs. 6000.0
Discount: Rs. 600.0
Total tax: Rs. 738.0
Final amount: Rs. 6138.0
```

Conclusion

The program generates an itemized retail bill with a threshold discount, a maximum discount limit, and tax rates based on product category.